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Intelligent actuated temple tips

Abstract

Systems and methods herein describe an actuator control system for controlling intelligent actuated temple tips of a pair of eyeglasses. The actuator control system receives a set of measurements corresponding to a length of eyeglass temple tips on a pair of eyeglasses, receives, from a user of the pair of eyeglasses, user feedback corresponding to a position of the eyeglass temple tips, generating a predicted set of measurements for the length of the eyeglass temple tips using a machine learning model, and transmits the predicted set of measurements to pneumatic actuators coupled to the eyeglass temple tips.

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Background/Summary

TECHNICAL FIELD
(1) Embodiments herein generally relate to adjusting eyeglass frame dimensions. More specifically, but not by way of limitation, embodiments herein describe an actuator control system for intelligent actuated temple tips.
BACKGROUND
(2) Eyeglass temples are the stems of the frames that connect the front of the glasses to the back of the wearer's head, just behind the wearer's ears. Eyeglass temple measurements contribute to the comfort of the glasses for a wearer. Eyeglass temple tips are the portion of the eyeglass temples that are placed around the wearer's ears.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
(1) In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some nonlimiting examples are illustrated in the figures of the accompanying drawings in which:
(2) FIG. **1** is a block diagram showing an example messaging system for exchanging data over a network, according to example embodiments.
(3) FIG. **2** illustrates a head-wearable apparatus, according to one example embodiment
(4) FIG. **3** is a flowchart of a method for remotely adjusting the length of the temple tips on a pair of wearable eyeglasses, according to example embodiments.
(5) FIG. **4** is a diagrammatic representation of a machine within which instructions for causing the machine to perform any one or more of the methodologies discussed herein may be executed.
(6) FIG. **5** is a block diagram illustrating a software architecture, which can be installed on any one or more of the devices described herein.
(7) FIG. **6** illustrates a system in which a head-wearable apparatus can be implemented according to one example embodiment.
DETAILED DESCRIPTION
(8) Traditionally eyeglass frames cannot be altered without a complete replacement of parts. Most portions of the eyeglass frames are not easily adjustable and replacing an eyeglass frame can be expensive. Specifically, temple tips on pair of eyeglasses are typically must be adjusted (e.g., bent or straightened), by applying heat. However, bending or straightening the temple tips does not change the length of the temples of the eyeglasses and only how the temple tips fit around a wearer's ears.
(9) Systems and methods herein describe intelligent actuated glasses' temple tips. The intelligent actuated glasses' temple tips are controlled by an actuator control system. The actuator control system adjusts the lengths of the temple tips of a pair of eyeglasses so as to provide a more comfortable fit to a wearer of the eyeglasses.
(10) The pair of eyeglasses may be an apparatus with at least one of augmented reality or virtual

reality capabilities. The pair of eyeglasses contains two actuators on each temple tip of the pair of eyeglasses. The actuators may be, for example, pneumatic actuators that are connected via an air line to a tank of compressed air or gas.

(11) The actuator control system receives a current length of the eyeglass temples tips on the pair of eyeglasses and feedback from the user that corresponds to the current fit of the eyeglasses. Based on the current length and the received user feedback, the actuator control system generates a predicted length for the temple tips. The predicted length is generated using a machine learning model. The machine learning model may be trained on historical user data. After generating the predicted lengths for the temple tips, the actuator control system transmits the predicted lengths to the pneumatic actuators. The actuator control system causes the pneumatic actuators to modify the current length of the temple tips based on the predicted lengths (e.g., either expand or contract the length of the temple tips). Further details regarding the actuator control system are described below.

(12) Networked Computing Environment

(13) FIG. **1** is a block diagram showing an example messaging system **100** for exchanging data (e.g., messages and associated content) over a network. The messaging system **100** includes multiple instances of a client device **102**, each of which hosts a number of applications, including a messaging client **104** and other applications **106**. Each messaging client **104** is communicatively coupled to other instances of the messaging client **104** (e.g., hosted on respective other client devices **102**), a messaging server system **108** and third-party servers **110** via a network **112** (e.g., the Internet). A messaging client **104** can also communicate with locally-hosted applications **106** using Applications Program Interfaces (APIs).

(14) A messaging client **104** is able to communicate and exchange data with other messaging clients **104** and with the messaging server system **108** via the network **112**. The data exchanged between messaging clients **104**, and between a messaging client **104** and the messaging server system **108**, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., text, audio, video or other multimedia data).

(15) The messaging server system **108** provides server-side functionality via the network **112** to a particular messaging client **104**. While certain functions of the messaging system **100** are described herein as being performed by either a messaging client **104** or by the messaging server system **108**, the location of certain functionality either within the messaging client **104** or the messaging server system **108** may be a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the messaging server system **108** but to later migrate this technology and functionality to the messaging client **104** where a client device **102** has sufficient processing capacity.

(16) The messaging server system **108** supports various services and operations that are provided to the messaging client **104**. Such operations include transmitting data to, receiving data from, and processing data generated by the messaging client **104**. This data may include message content, client device information, geolocation information, media augmentation and overlays, message content persistence conditions, social network information, and live event information, as examples. Data exchanges within the messaging system **100** are invoked and controlled through functions available via user interfaces (UIs) of the messaging client **104**.

(17) Turning now specifically to the messaging server system **108**, an Application Program Interface (API) server **116** is coupled to, and provides a programmatic interface to, application servers **114**. The application servers **114** are communicatively coupled to a database server **120**, which facilitates access to a database **126** that stores data associated with messages processed by the application servers **114**. Similarly, a web server **128** is coupled to the application servers **114**, and provides web-based interfaces to the application servers **114**. To this end, the web server **128** processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

(18) The Application Program Interface (API) server **116** receives and transmits message data (e.g.,

commands and message payloads) between the client device **102** and the application servers **114**. Specifically, the Application Program Interface (API) server **116** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the messaging client **104** in order to invoke functionality of the application servers **114**. The Application Program Interface (API) server **116** exposes various functions supported by the application servers **114**, including account registration, login functionality, the sending of messages, via the application servers **114**, from a particular messaging client **104** to another messaging client **104**, the sending of media files (e.g., images or video) from a messaging client **104** to a messaging server **118**, and for possible access by another messaging client **104**, the settings of a collection of media data (e.g., story), the retrieval of a list of friends of a user of a client device **102**, the retrieval of such collections, the retrieval of messages and content, the addition and deletion of entities (e.g., friends) to an entity graph (e.g., a social graph), the location of friends within a social graph, and opening an application event (e.g., relating to the messaging client **104**).
(19) The application servers **114** host a number of server applications and subsystems, including for example a messaging server **118**, an image processing server **122**, a social network server **124**, and an actuator control system **130**. The messaging server **118** implements a number of message processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., textual and multimedia content) included in messages received from multiple instances of the messaging client **104**. As will be described in further detail, the text and media content from multiple sources may be aggregated into collections of content (e.g., called stories or galleries). These collections are then made available to the messaging client **104**. Other processor and memory intensive processing of data may also be performed server-side by the messaging server **118**, in view of the hardware requirements for such processing.
(20) The application servers **114** also include an image processing server **122** that is dedicated to performing various image processing operations, typically with respect to images or video within the payload of a message sent from or received at the messaging server **118**.
(21) The social network server **124** supports various social networking functions and services and makes these functions and services available to the messaging server **118**. Examples of functions and services supported by the social network server **124** include the identification of other users of the messaging system **100** with which a particular user has relationships or is “following,” and also the identification of other entities and interests of a particular user.
(22) The actuator control system **130** generates adjustments to the length of an eyeglasses' temple tips by remotely controlling a pair of eyeglasses with an AR/VR display system. For example, the actuator control system **130** may intelligently predict the lengths of the temple tips for a unique user using a machine learning model and may adjust the lengths of the temples tips by shrinking or expanding the temples tips using a series of actuators.
(23) Returning to the messaging client **104**, features and functions of an external resource (e.g., an application **106** or applet) are made available to a user via an interface of the messaging client **104**. In this context, “external” refers to the fact that the application **106** or applet is external to the messaging client **104**. The external resource is often provided by a third party but may also be provided by the creator or provider of the messaging client **104**. The messaging client **104** receives a user selection of an option to launch or access features of such an external resource. The external resource may be the application **106** installed on the client device **102** (e.g., a “native app”), or a small-scale version of the application (e.g., an “applet”) that is hosted on the client device **102** or remote of the client device **102** (e.g., on third-party servers **110**). The small-scale version of the application includes a subset of features and functions of the application (e.g., the full-scale, native version of the application) and is implemented using a markup-language document. In one example, the small-scale version of the application (e.g., an “applet”) is a web-based, markup-language version of the application and is embedded in the messaging client **104**. In addition to using markup-language documents (e.g., a .*ml file), an applet may incorporate a scripting

language (e.g., a .*js file or a .json file) and a style sheet (e.g., a .*ss file).
(24) In response to receiving a user selection of the option to launch or access features of the external resource, the messaging client **104** determines whether the selected external resource is a web-based external resource or a locally-installed application **106**. In some cases, applications **106** that are locally installed on the client device **102** can be launched independently of and separately from the messaging client **104**, such as by selecting an icon, corresponding to the application **106**, on a home screen of the client device **102**. Small-scale versions of such applications can be launched or accessed via the messaging client **104** and, in some examples, no or limited portions of the small-scale application can be accessed outside of the messaging client **104**. The small-scale application can be launched by the messaging client **104** receiving, from a third-party server **110** for example, a markup-language document associated with the small-scale application and processing such a document.
(25) In response to determining that the external resource is a locally-installed application **106**, the messaging client **104** instructs the client device **102** to launch the external resource by executing locally-stored code corresponding to the external resource. In response to determining that the external resource is a web-based resource, the messaging client **104** communicates with the third-party servers **110** (for example) to obtain a markup-language document corresponding to the selected external resource. The messaging client **104** then processes the obtained markup-language document to present the web-based external resource within a user interface of the messaging client **104**.
(26) The messaging client **104** can notify a user of the client device **102**, or other users related to such a user (e.g., "friends"), of activity taking place in one or more external resources. For example, the messaging client **104** can provide participants in a conversation (e.g., a chat session) in the messaging client **104** with notifications relating to the current or recent use of an external resource by one or more members of a group of users. One or more users can be invited to join in an active external resource or to launch a recently-used but currently inactive (in the group of friends) external resource. The external resource can provide participants in a conversation, each using respective messaging clients **104**, with the ability to share an item, status, state, or location in an external resource with one or more members of a group of users into a chat session. The shared item may be an interactive chat card with which members of the chat can interact, for example, to launch the corresponding external resource, view specific information within the external resource, or take the member of the chat to a specific location or state within the external resource. Within a given external resource, response messages can be sent to users on the messaging client **104**. The external resource can selectively include different media items in the responses, based on a current context of the external resource.
(27) The messaging client **104** can present a list of the available external resources (e.g., applications **106** or applets) to a user to launch or access a given external resource. This list can be presented in a context-sensitive menu. For example, the icons representing different ones of the application **106** (or applets) can vary based on how the menu is launched by the user (e.g., from a conversation interface or from a non-conversation interface).
(28) FIG. **2** illustrates a head-wearable apparatus **200**, according to one example embodiment. FIG. **2** illustrates a perspective view of the head-wearable apparatus **200** according to one example embodiment. In some examples, the client device **102** may include the head-wearable apparatus **200**. In some examples, the head-wearable apparatus may be part of the messaging server system **108**.
(29) In FIG. **2**, the head-wearable apparatus **200** is a pair of eyeglasses. In some embodiments, the head-wearable apparatus **200** can be sunglasses or goggles. Some embodiments can include one or more wearable devices, such as a pendant with an integrated camera that is integrated with, in communication with, or coupled to, the head-wearable apparatus **200** or a client device **102**. Any desired wearable device may be used in conjunction with the embodiments of the present

disclosure, such as a watch, a headset, a wristband, earbuds, clothing (such as a hat or jacket with integrated electronics), a clip-on electronic device, or any other wearable devices. It is understood that, while not shown, one or more portions of the system included in the head-wearable apparatus **200** can be included in a client device **102** that can be used in conjunction with the head-wearable apparatus **200**.

(30) In FIG. **2**, the head-wearable apparatus **200** is a pair of eyeglasses that includes a frame **210** that includes eye wires (or rims) that are coupled to two stems (or temples), respectively, via hinges and/or end pieces. The eye wires of the frame **210** carry or hold a pair of lenses (e.g., lens **212** and lens **214**). The frame **210** includes a first (e.g., right) side that is coupled to the first stem and a second (e.g., left) side that is coupled to the second stem. The first side is opposite the second side of the frame **210**.

(31) The head-wearable apparatus **200** further includes a camera module (not shown) that includes camera lenses (e.g., camera lens **206**, camera lens **208**) and at least one image sensor. The camera lens **206** and camera lens **208** may be a perspective camera lens or a non-perspective camera lens. A non-perspective camera lens may be, for example, a fisheye lens, a wide-angle lens, an omnidirectional lens, etc. The image sensor captures digital video through the camera lens **206** and camera lens **208**. The images may be also be still image frame or a video including a plurality of still image frames. The camera module can be coupled to the frame **210**. As shown in FIG. **1** the frame **210** is coupled to the camera lens **206** and camera lens **208** such that the camera lenses (e.g., camera lens **206**, camera lens **208**) face away from the user (e.g., rear-facing cameras). The camera lens **206** and camera lens **208** can be perpendicular to the lens **212** and lens **214**. The camera module can further include dual-front facing cameras (e.g., cameras that face the user) that are separated by the width of the frame **210** or the width of the head of the user of the head-wearable apparatus **200**.

(32) In FIG. **2**, the two stems (or temples) are respectively coupled to microphone housing **202** and microphone housing **204**. The first and second stems are coupled to opposite sides of a frame **210** of the head-wearable apparatus **200**. The first stem is coupled to the first microphone housing **202** and the second stem is coupled to the second microphone housing **204**. The microphone housing **202** and microphone housing **204** can be coupled to the stems between the locations of the frame **210** and the temple tips (temple tip **216** and temple tip **218**). The microphone housing **202** and microphone housing **204** can be located on either side of the user's temples when the user is wearing the head-wearable apparatus **200**.

(33) The temple tip **216** includes a first actuator system and the and temple tip **218** includes a second actuator system. Each of the first actuator system and the second actuator system includes two pneumatic actuators. The pneumatic actuators of each actuator system are connected via one or more air lines to an air tank (not pictured). The air tank may be some form of pressurized gas. The first actuator system and second actuator system expand or shrink temple tip **216** and the temple tip **218** respectively, based on measurements provided by the actuator control system **130**. A first actuator in the first actuator system moves the temple tip **216** in a horizontal direction, while a second actuator in the first actuator system moves the temple tip **216** in a vertical direction. Similarly, a first actuator in the second actuator system moves the temple tip **218** in a horizontal direction, while a second actuator in the second actuator system moves the temple tip **218** in a vertical direction.

(34) As shown in FIG. **2**, the microphone housing **202** and microphone housing **204** encase a plurality of microphones (not shown). The microphones are air interface sound pickup devices that convert sound into an electrical signal. More specifically, the microphones are transducers that convert acoustic pressure into electrical signals (e.g., acoustic signals). Microphones can be digital or analog microelectro-mechanical systems (MEMS) microphones. The acoustic signals generated by the microphones can be pulse density modulation (PDM) signals.

(35) Although the described flow diagram below can show operations as a sequential process,

many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed. A process may correspond to a method, a procedure, an algorithm, etc. The operations of methods may be performed in whole or in part, may be performed in conjunction with some or all of the operations in other methods, and may be performed by any number of different systems, such as the systems described herein, or any portion thereof, such as a processor included in any of the systems.

(36) FIG. **3** is a method **300** for remotely adjusting the length of the temple tips on a pair of wearable eyeglasses, according to example embodiments. In one example, the processor in client device **102**, the processor in a head-wearable apparatus **200**, the processor in the messaging server system **108**, the processor in the actuator control system **130**, or any combination thereof, can perform the operations in the method **300**.

(37) At operation **302**, the actuator control system **130** receives a set of measurements corresponding to lengths of eyeglass temple tips on a pair of eyeglasses. The set of measurements may include a first length of the first temple tip (e.g., temple tip **216**) and a second length of the second temple tip (e.g., temple tip **218**). The measurements may be received by one or more sensors of the head-wearable apparatus **200**. In some examples, the actuator control system **130** further captures a first set of images of the wearer using front-facing cameras of the head-wearable apparatus.

(38) At operation **304**, the actuator control system **130** receives, from a user of the pair of eyeglasses, user feedback corresponding to a position of the eyeglass temple tips. The feedback may be received via input received on an AR/VR display screen of the head-wearable apparatus **200**. For example, the actuator control system **130** may cause display of one or more user interface elements (e.g., buttons, checkboxes, toggle switch) on the AR/VR display screen of the head-wearable apparatus **200**. In some examples, the feedback is audio feedback that received from one or more microphones (e.g., located within the microphone housing **202**, microphone housings **202**). In some examples, the feedback is hand gestures or poses (e.g., thumbs up, thumbs down, and the like) that is received from the camera system of the head-wearable apparatus **200** (e.g., rear-facing cameras).

(39) At operation **306**, based on the user feedback and the set of measurements, the actuator control system **130** generates a predicted set of measurements for the length of the eyeglass temple tips. For example, the actuator control system **130** may use a machine learning model that is trained to generate the predicted set of measurements. The machine learning model may be trained on historical user data that includes but is not limited to: eyeglass temple tip lengths, user face measurements, eyeglass lenses sizes, and historical user feedback. The machine learning model may use further data to generate the predicted set of measurements.

(40) At operation **308**, the actuator control system **130** transmits the predicted set of measurements to pneumatic actuators coupled to the eyeglass temple tips. The actuator control system **130** causes each set of pneumatic actuators to expand or shrink the materials of each of the temple tip **216** and temple tip **218** based on the predicted set of measurements. The temple tip **216** and temple tip **218** of the pair of eyeglasses may be constructed out of a material that can roll in and out to expand or shrink the required distance. The material may be inside a tubular cover and can move in and out of the exterior tubular cover to adjust its size. In some examples the material is a plastic or silicon.

(41) In some examples, the actuator control system **130** captures a second set of images of the wearer via front-facing cameras of the head-wearable apparatus. The actuator control system **130** may compare the first set of captured images and the second set of captured images. For example, actuator control system **130** determines the heart rate or blood pressure of the wearer using the captured images. The actuator control system **130** may further receive heart rate or blood pressure information from one or more non-camera sensors (e.g., capacitive sensors) of the head-wearable apparatus **200**. If the heart rate or blood pressure is higher than an average threshold, the actuator

control system **130** may determine that the temple tips should be expanded so as to lessen the pressure of the eyeglasses temple tips against the wearer.
(42) After comparing the first set of images with the second set of images, the actuator control system **130** generates an adjusted set of measurements based on the comparison of the images. The adjusted set of measurements are a modification to the predicted set of measurements. The actuator control system **130** transmits the adjusted set of measurements to both sets of pneumatic actuators coupled to each of the eyeglass temple tips (e.g., temple tip **216**, temple tip **218**). The actuator control system **130** may continue to receive feedback from the wearer of the head-wearable apparatus and adjust the lengths of the temple tip **216** and temple tip **218** until the wearer is comfortable with the fit.
(43) Machine Architecture
(44) FIG. **4** is a diagrammatic representation of the machine **400** within which instructions **410** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **400** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **410** may cause the machine **400** to execute any one or more of the methods described herein. The instructions **410** transform the general, non-programmed machine **400** into a particular machine **400** programmed to carry out the described and illustrated functions in the manner described. The machine **400** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **400** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **400** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **410**, sequentially or otherwise, that specify actions to be taken by the machine **400**. Further, while only a single machine **400** is illustrated, the term "machine" shall also be taken to include a collection of machines that individually or jointly execute the instructions **410** to perform any one or more of the methodologies discussed herein. The machine **400**, for example, may comprise the client device **102** or any one of a number of server devices forming part of the messaging server system **108**. In some examples, the machine **400** may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.
(45) The machine **400** may include processors **404**, memory **406**, and input/output I/O components **402**, which may be configured to communicate with each other via a bus **440**. In an example, the processors **404** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **408** and a processor **412** that execute the instructions **410**. The term "processor" is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as "cores") that may execute instructions contemporaneously. Although FIG. **4** shows multiple processors **404**, the machine **400** may include a single processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.
(46) The memory **406** includes a main memory **414**, a static memory **416**, and a storage unit **418**, both accessible to the processors **404** via the bus **440**. The main memory **406**, the static memory

416, and storage unit **418** store the instructions **410** embodying any one or more of the methodologies or functions described herein. The instructions **410** may also reside, completely or partially, within the main memory **414**, within the static memory **416**, within machine-readable medium **420** within the storage unit **418**, within at least one of the processors **404** (e.g., within the Processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **400**.

(47) The I/O components **402** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **402** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **402** may include many other components that are not shown in FIG. **4**. In various examples, the I/O components **402** may include user output components **426** and user input components **428**. The user output components **426** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components **428** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(48) In further examples, the I/O components **402** may include biometric components **430**, motion components **432**, environmental components **434**, or position components **436**, among a wide array of other components. For example, the biometric components **430** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components **432** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

(49) The environmental components **434** include, for example, one or cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

(50) With respect to cameras, the client device **102** may have a camera system comprising, for example, front cameras on a front surface of the client device **102** and rear cameras on a rear surface of the client device **102**. The front cameras may, for example, be used to capture still images and video of a user of the client device **102** (e.g., “selfies”), which may then be augmented with augmentation data (e.g., filters) described above. The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode, with these images similarly being augmented with augmentation data. In addition to front and rear cameras, the client device **102** may also include a 360° camera for capturing 360° photographs and videos.

(51) Further, the camera system of a client device **102** may include dual rear cameras (e.g., a primary camera as well as a depth-sensing camera), or even triple, quad or penta rear camera configurations on the front and rear sides of the client device **102**. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera and a depth sensor, for example.
(52) The position components **436** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.
(53) Communication may be implemented using a wide variety of technologies. The I/O components **402** further include communication components **438** operable to couple the machine **400** to a network **422** or devices **424** via respective coupling or connections. For example, the communication components **438** may include a network interface Component or another suitable device to interface with the network **422**. In further examples, the communication components **438** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **424** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).
(54) Moreover, the communication components **438** may detect identifiers or include components operable to detect identifiers. For example, the communication components **438** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **438**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.
(55) The various memories (e.g., main memory **414**, static memory **416**, and memory of the processors **404**) and storage unit **418** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **410**), when executed by processors **404**, cause various operations to implement the disclosed examples.
(56) The instructions **410** may be transmitted or received over the network **422**, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components **438**) and using any one of several well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **410** may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices **424**.
(57) Software Architecture
(58) FIG. **5** is a block diagram **500** illustrating a software architecture **504**, which can be installed on any one or more of the devices described herein. The software architecture **504** is supported by hardware such as a machine **502** that includes processors **520**, memory **526**, and I/O components **538**. In this example, the software architecture **504** can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture **504** includes layers such as an operating system **512**, libraries **510**, frameworks **508**, and applications **506**. Operationally, the applications **506** invoke API calls **550** through the software stack and receive messages **552** in response to the API calls **550**.

(59) The operating system **512** manages hardware resources and provides common services. The operating system **512** includes, for example, a kernel **514**, services **516**, and drivers **522**. The kernel **514** acts as an abstraction layer between the hardware and the other software layers. For example, the kernel **514** provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionality. The services **516** can provide other common services for the other software layers. The drivers **522** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **522** can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.
(60) The libraries **510** provide a common low-level infrastructure used by the applications **506**. The libraries **510** can include system libraries **518** (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **510** can include API libraries **524** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries **510** can also include a wide variety of other libraries **528** to provide many other APIs to the applications **506**.
(61) The frameworks **508** provide a common high-level infrastructure that is used by the applications **506**. For example, the frameworks **508** provide various graphical user interface (GUI) functions, high-level resource management, and high-level location services. The frameworks **508** can provide a broad spectrum of other APIs that can be used by the applications **506**, some of which may be specific to a particular operating system or platform.
(62) In an example, the applications **506** may include a home application **536**, a contacts application **530**, a browser application **532**, a book reader application **534**, a location application **542**, a media application **544**, a messaging application **546**, a game application **548**, and a broad assortment of other applications such as a third-party application **540**. The applications **506** are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications **506**, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the third-party application **540** (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the third-party application **540** can invoke the API calls **550** provided by the operating system **512** to facilitate functionality described herein.
(63) System with Head-Wearable Apparatus
(64) FIG. **6** illustrates a system in which the head-wearable apparatus **200** can be implemented according to one example embodiment. FIG. **6** is a high-level functional block diagram of an example head-wearable apparatus **200** communicatively coupled a mobile client device **102** (or client device **604**) and a server system **632** via various network **638**.
(65) Head-wearable apparatus **200** includes a camera, such as at least one of visible light camera **612**, infrared emitter **614** and infrared camera **616**. The camera can include the camera module with the camera lens **206** and camera lens **208** in FIG. **1**.

(66) Client device **102** can be capable of connecting with head-wearable apparatus **200** using both a low-power wireless connection **634** and a high-speed wireless connection **636**. Client device **102** is connected to server system **632** and network **638**. The network **638** may include any combination of wired and wireless connections.

(67) Head-wearable apparatus **200** further includes two image displays of the image display of optical assembly **602**. The two image displays image display of optical assembly **602** include one associated with the left lateral side and one associated with the right lateral side of the head-wearable apparatus **200**. Head-wearable apparatus **200** also includes image display driver **608**, image processor **610**, low-power low power circuitry **626**, and high-speed circuitry **618**. Image display of optical assembly **602** are for presenting images and videos, including an image that can include a graphical user interface to a user of the head-wearable apparatus **200**.

(68) Image display driver **608** commands and controls the image display of the image display of optical assembly **602**. Image display driver **608** may deliver image data directly to the image display of the image display of optical assembly **602** for presentation or may have to convert the image data into a signal or data format suitable for delivery to the image display device. For example, the image data may be video data formatted according to compression formats, such as H. 264 (MPEG-4 Part **10**), HEVC, Theora, Dirac, RealVideo RV40, VP8, VP9, or the like, and still image data may be formatted according to compression formats such as Portable Network Group (PNG), Joint Photographic Experts Group (JPEG), Tagged Image File Format (TIFF) or exchangeable image file format (Exif) or the like.

(69) As noted above, head-wearable apparatus **200** includes a frame **210** and stems (or temples) extending from a lateral side of the frame **210**. Head-wearable apparatus **200** further includes a user input device **606** (e.g., touch sensor or push button) including an input surface on the head-wearable apparatus **200**. The user input device **606** (e.g., touch sensor or push button) is to receive from the user an input selection to manipulate the graphical user interface of the presented image.

(70) The components shown in FIG. **6** for the head-wearable apparatus **200** are located on one or more circuit boards, for example a PCB or flexible PCB, in the rims or temples. Alternatively or additionally, the depicted components can be located in the chunks, frames, hinges, or bridge of the head-wearable apparatus **200**. Left and right visible light cameras **612** can include digital camera elements such as a complementary metal-oxide-semiconductor (CMOS) image sensor, charge coupled device, a camera lens **206** and camera lens **208**, or any other respective visible or light capturing elements that may be used to capture data, including images of scenes with unknown objects.

(71) Head-wearable apparatus **200** includes a memory **622** which stores instructions to perform a subset or all of the functions described herein. Memory **622** can also include storage device.

(72) As shown in FIG. **6**, high-speed circuitry **618** includes high-speed processor **620**, memory **622**, and high-speed wireless circuitry **624**. In the example, the image display driver **608** is coupled to the high-speed circuitry **618** and operated by the high-speed processor **620** in order to drive the left and right image displays of the image display of optical assembly **602**. High-speed processor **620** may be any processor capable of managing high-speed communications and operation of any general computing system needed for head-wearable apparatus **200**. High-speed processor **620** includes processing resources needed for managing high-speed data transfers on high-speed wireless connection **636** to a wireless local area network (WLAN) using high-speed wireless circuitry **624**. In certain examples, the high-speed processor **620** executes an operating system such as a LINUX operating system or other such operating system of the head-wearable apparatus **200** and the operating system is stored in memory **622** for execution. In addition to any other responsibilities, the high-speed processor **620** executing a software architecture for the head-wearable apparatus **200** is used to manage data transfers with high-speed wireless circuitry **624**. In certain examples, high-speed wireless circuitry **624** is configured to implement Institute of Electrical and Electronic Engineers (IEEE) 802.11 communication standards, also referred to

herein as Wi-Fi. In other examples, other high-speed communications standards may be implemented by high-speed wireless circuitry **624**.

(73) Low-power wireless circuitry **630** and the high-speed wireless circuitry **624** of the head-wearable apparatus **200** can include short range transceivers (Bluetooth™) and wireless wide, local, or wide area network transceivers (e.g., cellular or WiFi). Client device **102**, including the transceivers communicating via the low-power wireless connection **634** and high-speed wireless connection **636**, may be implemented using details of the architecture of the head-wearable apparatus **200**, as can other elements of network **638**.

(74) Memory **622** includes any storage device capable of storing various data and applications, including, among other things, camera data generated by the left and right visible light cameras **612**, infrared camera **616**, and the image processor **610**, as well as images generated for display by the image display driver **608** on the image displays of the image display of optical assembly **602**. While memory **622** is shown as integrated with high-speed circuitry **618**, in other examples, memory **622** may be an independent standalone element of the head-wearable apparatus **200**. In certain such examples, electrical routing lines may provide a connection through a chip that includes the high-speed processor **620** from the image processor **610** or low-power processor **628** to the memory **622**. In other examples, the high-speed processor **620** may manage addressing of memory **622** such that the low-power processor **628** will boot the high-speed processor **620** any time that a read or write operation involving memory **622** is needed.

(75) As shown in FIG. **6**, the low-power processor **628** or high-speed processor **620** of the head-wearable apparatus **200** can be coupled to the camera (visible light camera **612**; infrared emitter **614**, or infrared camera **616**), the image display driver **608**, the user input device **606** (e.g., touch sensor or push button), and the memory **622**.

(76) Head-wearable apparatus **200** is connected with a host computer. For example, the head-wearable apparatus **200** is paired with the client device **102** via the high-speed wireless connection **636** or connected to the server system **632** via the network **638**. Server system **632** may be one or more computing devices as part of a service or network computing system, for example, that include a processor, a memory, and network communication interface to communicate over the network **638** with the client device **102** and head-wearable apparatus **200**.

(77) The client device **102** includes a processor and a network communication interface coupled to the processor. The network communication interface allows for communication over the network **638**, low-power wireless connection **634** or high-speed wireless connection **636**. Client device **102** can further store at least portions of the instructions for generating a binaural audio content in the client device **102**'s memory to implement the functionality described herein.

(78) Output components of the head-wearable apparatus **200** include visual components, such as a display such as a liquid crystal display (LCD), a plasma display panel (PDP), a light emitting diode (LED) display, a projector, or a waveguide. The image displays of the optical assembly are driven by the image display driver **608**. The output components of the head-wearable apparatus **200** further include acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor), other signal generators, and so forth. The input components of the head-wearable apparatus **200**, the client device **102**, and server system **632**, such as the user input device **606**, may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instruments), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(79) Head-wearable apparatus **200** may optionally include additional peripheral device elements. Such peripheral device elements may include biometric sensors, additional sensors, or display elements integrated with head-wearable apparatus **200**. For example, peripheral device elements

may include any I/O components including output components, motion components, position components, or any other such elements described herein.
(80) For example, the biometric components include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The position components include location sensor components to generate location coordinates (e.g., a Global Positioning System (GPS) receiver component), WiFi or Bluetooth™ transceivers to generate positioning system coordinates, altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like. Such positioning system coordinates can also be received over low-power wireless connections **634** and high-speed wireless connection **636** from the client device **102** via the low-power wireless circuitry **630** or high-speed wireless circuitry **624**.
(81) Where a phrase similar to “at least one of A, B, or C,” “at least one of A, B, and C,” “one or more A, B, or C,” or “one or more of A, B, and C” is used, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C.
(82) Changes and modifications may be made to the disclosed embodiments without departing from the scope of the present disclosure. These and other changes or modifications are intended to be included within the scope of the present disclosure, as expressed in the following claims.
Glossary
(83) “Carrier signal” refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.
(84) “Client device” refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.
(85) “Communication network” refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP)

including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

(86) “Component” refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and

store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors **1004** or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.

(87) “Computer-readable storage medium” refers to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms “machine-readable medium,” “computer-readable medium” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure.

(88) “Ephemeral message” refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

(89) “Machine storage medium” refers to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks The terms “machine-storage medium,” “device-storage medium,” “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure. The terms “machine-storage media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium.”

(90) “Non-transitory computer-readable storage medium” refers to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine.

(91) “Signal medium” refers to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications

signals or other intangible media to facilitate communication of software or data. The term “signal medium” shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a matter as to encode information in the signal. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure.

Claims

1. A method for adjusting lengths of temple tips, the method performed by a hardware processor, the method comprising: receiving a set of measurements corresponding to lengths of eyeglass temple tips on a pair of eyeglasses; receiving, from a user of the pair of eyeglasses, user feedback corresponding to a position of the eyeglass temple tips; based on the user feedback and the set of measurements, generating a predicted set of measurements for the length of the eyeglass temple tips; and transmitting the predicted set of measurements to pneumatic actuators coupled to the eyeglass temple tips causing the pneumatic actuators to adjust the lengths of the temple tips.
2. The method of claim 1 wherein the predicted set of measurements are generated using a machine learning model trained on historical user data, the historical user data comprising at least one of: eyeglass temple tip lengths, user face measurements, and user feedback.
3. The method of claim 1, further comprising: receiving a first set of images from a first camera coupled to the pair of eyeglasses, the first set of images comprising a first view of the user wearing the pair of eyeglasses.
4. The method of claim 3, wherein the predicted set of measurements are further generated based on the first set of images.
5. The method of claim 3, wherein the first camera is a front facing camera that faces the wearer of the pair of eyeglasses.
6. The method of claim 3, wherein the first set of images are captured before transmission of the predicted set of measurements to the pair of pneumatic actuators, the method further comprising: receiving a second set of images from the first camera, the second set of images captured after transmission of the predicted set of measurements to the pair of pneumatic actuators; comparing the first set of images to the second set of images; based on the comparison, generating an adjusted set of measurements, the adjusted set of measurements comprising an adjustment to the predicted set of measurements; and transmitting the adjusted set of measurements to the pneumatic actuators.
7. The method of claim 1, wherein the eyeglass temple tips are a pair of eyeglass temple tips, and wherein a first eyeglass temple tip of the pair of eyeglass temples comprises a first actuator system and a second eyeglass temple tip of the pair of eyeglass temples comprises a second actuator system.
8. The method of claim 7, wherein the first actuator system and the second actuator system are physically coupled to an air tank via an air line.
9. The method of claim 7, wherein the first actuator system comprises a first pair of pneumatic actuators and the second actuator system comprises a second pair of pneumatic actuators.
10. The method of claim 1, wherein transmitting the predicted set of measurements to the pneumatic actuators coupled to the eyeglass temple tips, causes the pneumatic actuators to expand or shrink the eyeglass temple tips.
11. An apparatus for adjusting lengths of temple tips, the apparatus comprising: a pair of eyeglasses comprising a frame, the frame comprising a pair of lenses, a pair of temples, and a pair of actuated temple tips; a first actuator system coupled to a first actuated temple tip in the pair of actuated temple tips configured to adjust the length of the first actuated temple tip based on a first predicted set of measurements; a second actuator system coupled to a second actuated temple tip in the pair of actuated temple tips configured to adjust the length of the first actuated temple tip based on a first predicted set of measurements; and a camera system coupled to the frame of the pair of

eyeglasses, the camera system comprising a front facing camera and a rear-facing camera.

12. The apparatus of claim 11, wherein the first actuator system comprises a first pair of pneumatic actuators and the second actuator system comprises a second pair of pneumatic actuators.

13. A non-transitory computer-readable storage medium for adjusting lengths of temple tips, the computer-readable storage medium including instructions that when executed by a computer, cause the computer to: receive a set of measurements corresponding to lengths of eyeglass temple tips on a pair of eyeglasses; receive, from a user of the pair of eyeglasses, user feedback corresponding to a position of the eyeglass temple tips; based on the user feedback and the set of measurements, generate a predicted set of measurements for the length of the eyeglass temple tips; and transmit the predicted set of measurements to pneumatic actuators coupled to the eyeglass temple tips causing the pneumatic actuators to adjust the lengths of the temple tips.

14. The computer-readable storage medium of claim 13 wherein the predicted set of measurements are generated use a machine learning model trained on historical user data, the historical user data comprising at least one of: eyeglass temple tip lengths, user face measurements, and user feedback.

15. The computer-readable storage medium of claim 13, wherein the instructions further configure the computer to: receive a first set of images from a first camera coupled to the pair of eyeglasses, the first set of images comprising a first view of the user wearing the pair of eyeglasses.

16. The computer-readable storage medium of claim 15, wherein the predicted set of measurements are further generated based on the first set of images.

17. The computer-readable storage medium of claim 15, wherein the first camera is a front face camera that faces the user of the pair of eyeglasses.

18. The computer-readable storage medium of claim 15, wherein the first set of images are captured before transmission of the predicted set of measurements to the pair of pneumatic actuators, wherein the instructions further configure the computer to: receive a second set of images from the first camera, the second set of images captured after transmission of the predicted set of measurements to the pair of pneumatic actuators; compare the first set of images to the second set of images; based on the comparison, generate an adjusted set of measurements, the adjusted set of measurements comprising an adjustment to the predicted set of measurements; and transmit the adjusted set of measurements to the pneumatic actuators.

19. The computer-readable storage medium of claim 13, wherein the eyeglass temple tips are a pair of eyeglass temple tips, and wherein a first eyeglass temple tip of the pair of eyeglass temples comprises a first actuator system and a second eyeglass temple tip of the pair of eyeglass temples comprises a second actuator system.

20. The computer-readable storage medium of claim 19, wherein the first actuator system and the second actuator system are physically coupled to an air tank via an air line.